

Assignment E

Common Notations

We use the following notation:

S_t	:	Stock price at time t
σ	:	Volatility of the Stock (assumed constant)
r	:	Interest rate
T	:	Time to option expiration (in years)
K	:	Strike price
W_t	:	Brownian motion process
$N(0, 1)$	:	Standard normal distribution
$(A - B)^+$	:	$\text{Max}(A-B, 0)$

Assignment E (C/C++)

- ▶ Due: April 7 by 2 PM.
- ▶ Write a function to price European Call options using Monte Carlo.
- ▶ Use the function you wrote to price 1 million options.
- ▶ You may use the function shown below to generate input values randomly:
- ▶ Measure time taken to price 1 million options.
- ▶ Vectorizing/parallelizing code is NOT required.

Monte Carlo Technique: Background

- ▶ The (time 0) value of an option is the discounted *expectation* of its payoff under the risk-neutral measure¹.
- ▶ That means: to price an option we need to compute an *expected value*, i.e. an integral.
- ▶ Monte Carlo is a *numerical technique* used to **estimate** an integral.

¹http://en.wikipedia.org/wiki/Risk-neutral_measure

Stock Price SDE

- ▶ We assume a stock price process follows a *Geometric Brownian Motion*².
- ▶ Under the *risk neutral measure* the process of the stock price is given by the SDE:

$$dS_t = rS_t dt + \sigma S_t dW_t \quad (1)$$

where,

$$dW_t = N(0, dt) \quad (2)$$

$$W_t = \sqrt{t}N(0, 1) \quad \because W_0 = 0 \quad (3)$$

I.e. The Brownian motion has a normal distribution with mean zero (0) and variance t.

- ▶ Using the above SDE we get:

$$S_t = S_0 \exp((r - \sigma^2/2)t + \sigma\sqrt{t}N(0,1)) \quad (4)$$

²http://en.wikipedia.org/wiki/Geometric_Brownian_motion

- ▶ Using equation (4), we can simulate the stock price at any time t , in the future.
- ▶ $N(0, 1)$ is a random observation (random number) drawn from the standard normal distribution.
- ▶ Let's say, $z =$ a random number drawn from $N(0, 1)$.
- ▶ The stock price at the expiration (time T) is given by:

$$S_T = S_0 \exp((r - \sigma^2/2)T + \sigma z \sqrt{T}) \quad (5)$$

- ▶ That means, we can simulate the terminal stock price S_T (i.e. price at time T).
- ▶ What do we need to find the price of a European style option?

Monte Carlo Option Pricer: Step by Step

- ▶ Draw a random number (z_i) from the standard normal distribution, $N(0, 1)$. Here, the subscript i denotes the i^{th} draw.
- ▶ Simulate the stock price at the option expiration (S_T). The stock price at the expiration is given by:

$$S_{T,i} = S_0 \exp((r - \sigma^2/2)T + \sigma z_i \sqrt{T}) \quad (6)$$

- ▶ For a given price path (i^{th} realization), the European call payoff at time zero is given by:

$$C_i = \exp^{-rT} (S_{T,i} - K)^+ \quad (7)$$

- ▶ Repeat above steps for (M) trials, and calculate the option payoff for each path.
- ▶ Find the mean value of the option payoffs:

$$\hat{C} = \frac{\exp^{-rT}}{M} \sum_{i=1}^M (S_{T,i} - K)^+ \quad (8)$$

- ▶ This is an **estimate** of the option price.
- ▶ *The law of large numbers* states that such an estimate converges to the correct value as the number of trials (M) increases.